

First Passage Time of a 1D Biased Random Walk:

Exhaustive Derivation for Reflecting and Absorbing Boundaries with Environmental Disorder

Abstract

This document provides an exhaustively detailed, step-by-step algebraic derivation of the First Passage Time (FPT) distribution for a discrete 1D biased random walk bounded by a reflecting wall at $x = 0$ and an absorbing cliff at $x = N$. We explicitly derive the Master Equation from first principles by marginalizing over a random environment of obstacles. Every algebraic manipulation, trigonometric identity, and linear algebra concept is broken down to its most fundamental components.

1 The Physical Environment and the Annealed Master Equation

1.1 The Lattice and the Obstacles

Consider a particle moving on a 1D discrete lattice. The valid sites where the particle can exist are $x \in \{0, 1, 2, \dots, N - 1\}$. The environment contains random obstacles. Let $O \in \{0, 1\}$ represent the state of the *target* site for any given step:

- $O = 1$ represents a **blocked** site (obstacle present).
- $O = 0$ represents a **free** site (no obstacle).

We assume the obstacles are distributed independently with a constant spatial density ρ . Therefore, the probability of the target site being blocked is $P(O = 1) = \rho$, and the probability of it being free is $P(O = 0) = 1 - \rho$.

1.2 Intrinsic vs. Effective Probabilities

In a completely free environment (no obstacles), the particle has **intrinsic** transition probabilities:

- p_0 : Intrinsic probability of attempting to hop right.
- q_0 : Intrinsic probability of attempting to hop left.

We assume the particle always attempts to move, so $p_0 + q_0 = 1$. If the particle attempts to hop into a blocked site ($O = 1$), the move fails and the particle stays in its current position.

1.3 Marginalizing the Environment (The Annealed Approximation)

Instead of tracking the exact configuration of every obstacle, we average (marginalize) over the environment to find the **effective** transition probabilities (p, q, r) for a single step. By the Law of Total Probability:

1. Effective Right-Hop Probability (p):

$$p = \sum_{O \in \{0,1\}} P(\text{right}|O)P(O) \quad (1)$$

$$= P(\text{right}|O=0)P(O=0) + P(\text{right}|O=1)P(O=1) \quad (2)$$

$$= p_0(1 - \rho) + 0 \cdot \rho \quad (3)$$

$$= p_0(1 - \rho) \quad (4)$$

2. Effective Left-Hop Probability (q):

$$q = \sum_{O \in \{0,1\}} P(\text{left}|O)P(O) \quad (5)$$

$$= P(\text{left}|O=0)P(O=0) + P(\text{left}|O=1)P(O=1) \quad (6)$$

$$= q_0(1 - \rho) + 0 \cdot \rho \quad (7)$$

$$= q_0(1 - \rho) \quad (8)$$

3. Effective Stay Probability (r): The particle effectively stays in place if it attempts to hop into a blocked obstacle:

$$r = 1 - p - q \quad (9)$$

$$= 1 - p_0(1 - \rho) - q_0(1 - \rho) \quad (10)$$

$$= 1 - (p_0 + q_0)(1 - \rho) \quad (11)$$

$$= 1 - (1)(1 - \rho) = \rho \quad (12)$$

Intuition 1. *This is a profound physical result. The effective "stay" probability r is exactly equal to the obstacle density ρ . The obstacles do not change the ratio of right-to-left spatial bias (since $p/q = p_0/q_0$), but they act as a universal "temporal friction," forcing the particle to pause in place with probability ρ at every step.*

2 The Master Equations and Boundary Conditions

Let $P_t(x)$ be the probability that the particle is at site x at time t and has *not yet been absorbed*. We must define how this probability updates from time t to time $t+1$ for three distinct regions: the bulk, the left wall, and the right wall.

2.1 The Bulk Master Equation ($1 \leq x \leq N-2$)

For any site safely away from the boundaries, the particle can arrive from the left (hopping right), from the right (hopping left), or it can be a particle that was already at x and stayed in place due to an obstacle:

$$P_{t+1}(x) = pP_t(x-1) + qP_t(x+1) + rP_t(x). \quad (13)$$

2.2 The Left Boundary: Reflecting Wall at $x = 0$

At $x = 0$, there is a solid wall. The particle cannot move to $x = -1$. If a particle is at $x = 0$ and attempts to hop left (probability q), it bounces off the wall and remains at $x = 0$. If it encounters an obstacle (probability r), it also stays at $x = 0$. If a particle is at $x = 1$ and hops left (probability q), it arrives at $x = 0$. Therefore, the exact update rule for the left boundary is:

$$P_{t+1}(0) = qP_t(0) + rP_t(0) + qP_t(1) = (q + r)P_t(0) + qP_t(1). \quad (14)$$

2.3 The Right Boundary: Absorbing Cliff at $x = N$

At $x = N$, there is a cliff. The moment the particle reaches this site, it is permanently removed from the system. Therefore, the probability of finding a surviving particle at $x = N$ is strictly zero for all time:

$$P_t(N) = 0 \quad \text{for all } t \geq 1. \quad (15)$$

3 Solving the Bulk Equation: Separation of Variables

Because the bulk master equation is linear and the probabilities p, q , and r do not change over time, we can use the method of separation of variables. We guess that the solution can be factored into a purely spatial part and a purely temporal part:

$$P_t(x) = \phi(x)\mu^t. \quad (16)$$

Here, $\phi(x)$ is the spatial eigenfunction (the shape of the wave), and μ is the temporal eigenvalue (the decay rate per time step).

Algebraic Step 1 (Substitution into the Bulk Equation). *Substitute $P_t(x) = \phi(x)\mu^t$ into the bulk master equation $P_{t+1}(x) = pP_t(x-1) + qP_t(x+1) + rP_t(x)$:*

$$\phi(x)\mu^{t+1} = p(\phi(x-1)\mu^t) + q(\phi(x+1)\mu^t) + r(\phi(x)\mu^t). \quad (17)$$

Algebraic Step 2 (Factoring out the temporal component). *Notice that μ^{t+1} can be written as $\mu \cdot \mu^t$. On the right side, μ^t is a common factor in all three terms. Let us factor it out:*

$$\phi(x) \cdot \mu \cdot \mu^t = \mu^t [p\phi(x-1) + q\phi(x+1) + r\phi(x)]. \quad (18)$$

Algebraic Step 3 (Isolating the spatial equation). *Assuming $\mu \neq 0$ and $\mu^t \neq 0$, we can divide both sides of the equation by μ^t . This completely eliminates time from the equation:*

$$\mu\phi(x) = p\phi(x-1) + q\phi(x+1) + r\phi(x). \quad (19)$$

Algebraic Step 4 (Grouping the spatial terms). *To form a standard difference equation, we move the $r\phi(x)$ term from the right side to the left side by subtracting it from both sides:*

$$\mu\phi(x) - r\phi(x) = p\phi(x-1) + q\phi(x+1). \quad (20)$$

Factor out $\phi(x)$ on the left side:

$$(\mu - r)\phi(x) = p\phi(x-1) + q\phi(x+1). \quad (21)$$

Intuition 2. Look closely at this final spatial equation. It is exactly the same spatial eigenvalue equation as a random walk without obstacles, but with the eigenvalue μ replaced by $(\mu - r)$. This proves mathematically that the obstacles (represented by r) do not change the spatial shape of the probability waves at all; they merely shift the temporal decay rate uniformly.

4 The Characteristic Equation

To solve the spatial difference equation $(\mu - r)\phi(x) = p\phi(x - 1) + q\phi(x + 1)$, we use the standard exponential ansatz for linear recurrences. We assume the spatial shape takes the form of a geometric sequence:

$$\phi(x) = \lambda^x, \quad (22)$$

where λ is a constant base that we need to find.

Algebraic Step 5 (Substituting the spatial ansatz). Substitute $\phi(x) = \lambda^x$, $\phi(x - 1) = \lambda^{x-1}$, and $\phi(x + 1) = \lambda^{x+1}$ into the spatial equation:

$$(\mu - r)\lambda^x = p\lambda^{x-1} + q\lambda^{x+1}. \quad (23)$$

Algebraic Step 6 (Clearing the lowest power). To simplify this into a standard polynomial, we want the lowest power of λ to be $\lambda^0 = 1$. The lowest power currently present is λ^{x-1} . We divide every single term in the equation by λ^{x-1} :

$$\frac{(\mu - r)\lambda^x}{\lambda^{x-1}} = \frac{p\lambda^{x-1}}{\lambda^{x-1}} + \frac{q\lambda^{x+1}}{\lambda^{x-1}}. \quad (24)$$

Using the exponent rule $\frac{a^m}{a^n} = a^{m-n}$, this simplifies to:

$$(\mu - r)\lambda^{x-(x-1)} = p\lambda^{(x-1)-(x-1)} + q\lambda^{(x+1)-(x-1)}. \quad (25)$$

$$(\mu - r)\lambda^1 = p\lambda^0 + q\lambda^2. \quad (26)$$

Since $\lambda^1 = \lambda$ and $\lambda^0 = 1$, we get:

$$(\mu - r)\lambda = p + q\lambda^2. \quad (27)$$

Algebraic Step 7 (Forming the standard quadratic equation). We want to write this in the standard quadratic form $a\lambda^2 + b\lambda + c = 0$. Subtract $(\mu - r)\lambda$ from both sides:

$$q\lambda^2 - (\mu - r)\lambda + p = 0. \quad (28)$$

This is the **Characteristic Equation**.

4.1 Vieta's Formulas

Before we solve for λ explicitly, we can extract vital relationships between the roots using Vieta's formulas for a quadratic equation $a\lambda^2 + b\lambda + c = 0$. Here, $a = q$, $b = -(\mu - r)$, and $c = p$.

Let the two roots be λ_+ and λ_- . **The Product of the Roots:**

$$\lambda_+\lambda_- = \frac{c}{a} = \frac{p}{q}. \quad (29)$$

Recall that $p = p_0(1 - \rho)$ and $q = q_0(1 - \rho)$. Substituting these in:

$$\lambda_+\lambda_- = \frac{p_0(1 - \rho)}{q_0(1 - \rho)} = \frac{p_0}{q_0}. \quad (30)$$

Intuition 3. *The obstacle density ρ perfectly cancels out in the product of the roots! This mathematically proves that while obstacles slow down the particle temporally, they do not alter the fundamental spatial bias (the ratio of right-to-left drift) of the random walk.*

The Sum of the Roots:

$$\lambda_+ + \lambda_- = -\frac{b}{a} = -\frac{-(\mu - r)}{q} = \frac{\mu - r}{q}. \quad (31)$$

Because it is a second-order equation, the most general spatial solution is a linear combination of both roots:

$$\phi(x) = A\lambda_+^x + B\lambda_-^x, \quad (32)$$

where A and B are constants that will be locked in by the boundary conditions.

5 Polar Parametrization of the Roots

5.1 The Notational Trap: r vs R

Before we proceed, we must address a critical notational collision. In Section 1, we defined r as the effective probability of the particle staying in place due to obstacles ($r = \rho$). However, in standard complex analysis, the magnitude (radius) of a polar complex number is traditionally denoted by r . To prevent catastrophic confusion, we will strictly use the capital letter R to denote the magnitude of our complex roots, and keep the lowercase r exclusively for the obstacle "stay" probability.

5.2 The Need for Complex Numbers

Our characteristic equation is $q\lambda^2 - (\mu - r)\lambda + p = 0$. Let us define a "shifted" eigenvalue $\tilde{\mu} = \mu - r$. The equation becomes $q\lambda^2 - \tilde{\mu}\lambda + p = 0$. The roots are given by the quadratic formula:

$$\lambda_{\pm} = \frac{\tilde{\mu} \pm \sqrt{\tilde{\mu}^2 - 4pq}}{2q}. \quad (33)$$

Because the absorbing boundary at $x = N$ constantly drains probability from the system, the total probability must decay over time. This requires the true eigenvalue to be bounded ($|\mu| < 1$), which in turn restricts the shifted eigenvalue $\tilde{\mu}$. We know that the maximum possible value of $4pq$ is 1 (when $p = q = 0.5$). For decaying modes, it is almost always true that $\tilde{\mu}^2 < 4pq$. This makes the term inside the square root (the discriminant) strictly negative. A negative discriminant means the roots $\lambda_{\pm}$ are **complex conjugates**.

Intuition 4. *In linear algebra, real eigenvalues correspond to pure exponential growth or decay. Complex eigenvalues correspond to rotation and oscillation. Because the particle is "trapped" between a wall and a cliff, its probability wave must oscillate (to create nodes at the boundaries) while simultaneously decaying in amplitude. Complex roots perfectly capture this dual wave-like behavior.*

5.3 Transforming to Polar Coordinates

Working with the square root of a negative number in Cartesian form ($a \pm bi$) makes applying boundary conditions algebraically disastrous, especially when raising the roots

to the power of N . Instead, we map the complex roots to polar coordinates. Any complex number can be written as $Re^{i\theta}$, where R is the magnitude (radius) and θ is the phase angle. Because the roots of a polynomial with real coefficients must be complex conjugates, we define:

$$\lambda_+ = Re^{i\theta} \quad \text{and} \quad \lambda_- = Re^{-i\theta}. \quad (34)$$

Algebraic Step 8 (Finding the Magnitude R). *We use Vieta's product formula from Section 3: $\lambda_+\lambda_- = \frac{p}{q}$. Substitute the polar forms into the left side:*

$$(Re^{i\theta})(Re^{-i\theta}) = R^2e^{i\theta-i\theta} = R^2e^0 = R^2. \quad (35)$$

Equating the two expressions for the product:

$$R^2 = \frac{p}{q} \implies R = \sqrt{\frac{p}{q}}. \quad (36)$$

Thus, the exact polar forms of the roots are:

$$\lambda_{\pm} = \sqrt{\frac{p}{q}}e^{\pm i\theta}. \quad (37)$$

Algebraic Step 9 (Finding the Eigenvalue μ in terms of θ). *We use Vieta's sum formula: $\lambda_+ + \lambda_- = \frac{\tilde{\mu}}{q}$. Multiply by q to isolate $\tilde{\mu}$:*

$$\tilde{\mu} = q(\lambda_+ + \lambda_-). \quad (38)$$

Substitute the polar forms:

$$\tilde{\mu} = q \left(\sqrt{\frac{p}{q}}e^{i\theta} + \sqrt{\frac{p}{q}}e^{-i\theta} \right). \quad (39)$$

Factor out the magnitude $\sqrt{p/q}$:

$$\tilde{\mu} = q\sqrt{\frac{p}{q}}(e^{i\theta} + e^{-i\theta}). \quad (40)$$

Simplify the leading coefficient: $q\sqrt{p/q} = \sqrt{q^2 \cdot p/q} = \sqrt{pq}$.

$$\tilde{\mu} = \sqrt{pq}(e^{i\theta} + e^{-i\theta}). \quad (41)$$

Apply Euler's trigonometric identity, which states that $e^{i\theta} + e^{-i\theta} = 2\cos\theta$:

$$\tilde{\mu} = 2\sqrt{pq}\cos\theta. \quad (42)$$

Finally, recall that $\tilde{\mu}$ was just a shifted variable defined as $\tilde{\mu} = \mu - r$. We add r back to both sides to find the true temporal eigenvalue μ :

$$\mu = 2\sqrt{pq}\cos\theta + r. \quad (43)$$

Intuition 5. *This is a profound physical result. The temporal decay rate μ consists of two distinct parts: 1) $2\sqrt{pq}\cos\theta$: The spatial oscillation and base decay dictated by the left/right hopping bias. 2) $+r$: A uniform upward shift caused by the obstacles. Because $r = \rho$ (the obstacle density), the obstacles literally just "pause" the particle, artificially inflating the survival probability at each step by exactly the density of the environment. The spatial shape of the wave remains entirely unaffected!*

6 Applying the Absorbing Boundary Condition

We now have the general spatial solution $\phi(x) = A\lambda_+^x + B\lambda_-^x$. We must enforce the boundary conditions to find the specific shape of the wave. We start with the easier boundary: the absorbing cliff at $x = N$.

6.1 The Absorbing Cliff at $x = N$

The physical rule is that the probability of finding a surviving particle at the cliff is strictly zero:

$$\phi(N) = 0. \quad (44)$$

Algebraic Step 10 (Substituting the general solution). *Plug $x = N$ into the general solution:*

$$A\lambda_+^N + B\lambda_-^N = 0. \quad (45)$$

Algebraic Step 11 (Solving for the ratio A/B). *Move the A term to the right side:*

$$B\lambda_-^N = -A\lambda_+^N. \quad (46)$$

Divide both sides by B and by λ_+^N :

$$\frac{A}{B} = -\frac{\lambda_-^N}{\lambda_+^N} = -\left(\frac{\lambda_-}{\lambda_+}\right)^N. \quad (47)$$

Algebraic Step 12 (Substituting A back into the general solution). *Instead of keeping A and B as separate unknowns, we can express A entirely in terms of B :*

$$A = -B\frac{\lambda_-^N}{\lambda_+^N}. \quad (48)$$

Substitute this expression for A directly into $\phi(x) = A\lambda_+^x + B\lambda_-^x$:

$$\phi(x) = \left(-B\frac{\lambda_-^N}{\lambda_+^N}\right)\lambda_+^x + B\lambda_-^x. \quad (49)$$

Algebraic Step 13 (Factoring out common terms). *Factor out the common constant $B\lambda_-^N$ from both terms. To do this cleanly, rewrite the first term as $-B\lambda_-^N\left(\frac{\lambda_+^x}{\lambda_+^N}\right)$ and the second term as $B\lambda_-^N\left(\frac{\lambda_-^x}{\lambda_-^N}\right)$:*

$$\phi(x) = B\lambda_-^N \left[-\frac{\lambda_+^x}{\lambda_+^N} + \frac{\lambda_-^x}{\lambda_-^N} \right]. \quad (50)$$

Using the exponent rule $\frac{a^m}{a^n} = a^{m-n}$, this simplifies to:

$$\phi(x) = B\lambda_-^N \left[-\lambda_+^{x-N} + \lambda_-^{x-N} \right]. \quad (51)$$

Algebraic Step 14 (Changing variables for spatial symmetry). *Notice that the exponent is $x - N$. Since the particle is confined to $x < N$, this exponent is always negative. It is much more intuitive to measure the distance from the cliff. Let us define the distance to the cliff as $y = N - x$. Therefore, $x - N = -y$. Substitute $-y$ into the exponents:*

$$\phi(x) = B\lambda_-^N \left[-\lambda_+^{-y} + \lambda_-^{-y} \right] = B\lambda_-^N \left[\lambda_-^{-y} - \lambda_+^{-y} \right]. \quad (52)$$

Algebraic Step 15 (Substituting the Polar Forms). Now, substitute $\lambda_+ = Re^{i\theta}$ and $\lambda_- = Re^{-i\theta}$ into the bracketed term:

$$\lambda_-^{-y} = (Re^{-i\theta})^{-y} = R^{-y}e^{iy\theta}, \quad (53)$$

$$\lambda_+^{-y} = (Re^{i\theta})^{-y} = R^{-y}e^{-iy\theta}. \quad (54)$$

Plug these back into the bracket:

$$\phi(x) = B\lambda_-^N [R^{-y}e^{iy\theta} - R^{-y}e^{-iy\theta}]. \quad (55)$$

Factor out R^{-y} :

$$\phi(x) = B\lambda_-^N R^{-y} [e^{iy\theta} - e^{-iy\theta}]. \quad (56)$$

Algebraic Step 16 (Applying Euler's Sine Identity). Euler's formula states that $\sin(u) = \frac{e^{iu} - e^{-iu}}{2i}$, which means $e^{iu} - e^{-iu} = 2i \sin(u)$. Let $u = y\theta$. Applying this identity to our bracket:

$$\phi(x) = B\lambda_-^N R^{-y} [2i \sin(y\theta)]. \quad (57)$$

Algebraic Step 17 (Dropping irrelevant constants and reverting to x). Because eigenfunctions define the shape of the wave, any constant multiplier can be dropped (it will be reabsorbed by the expansion coefficients c_n later). The terms B , λ_-^N , and $2i$ are all independent of the spatial coordinate x . We drop them entirely. We are left with:

$$\phi(x) \propto R^{-y} \sin(y\theta). \quad (58)$$

Now, substitute $y = N - x$ back into the equation:

$$\phi(x) \propto R^{-(N-x)} \sin((N-x)\theta). \quad (59)$$

Using exponent rules, $R^{-(N-x)} = R^{x-N} = R^x \cdot R^{-N}$. Since R^{-N} is just another constant, we drop it too! We are left with the beautifully simple spatial eigenfunction:

$$\phi(x) = R^x \sin((N-x)\theta). \quad (60)$$

Finally, substitute $R = \sqrt{p/q} = (p/q)^{1/2}$:

$$\phi(x) = \left(\frac{p}{q}\right)^{x/2} \sin((N-x)\theta). \quad (61)$$

Intuition 6. Look at this final shape. The term $(p/q)^{x/2}$ is the spatial "tilt" caused by the bias (if $p > q$, the wave grows exponentially to the right). The term $\sin((N-x)\theta)$ is the oscillating wave. Notice that when $x = N$, the argument of the sine is $\sin(0) = 0$. The wave perfectly anchors itself to zero exactly at the absorbing cliff, satisfying the boundary condition flawlessly.

7 Applying the Reflecting Boundary Condition

We now possess the general spatial eigenfunction anchored to the absorbing cliff at $x = N$:

$$\phi(x) = \left(\frac{p}{q}\right)^{x/2} \sin((N-x)\theta). \quad (62)$$

We must now force this wave to satisfy the physical rules of the reflecting wall at $x = 0$.

7.1 The Master Equation at $x = 0$

Recall from Section 2 that the exact update rule for the left boundary, accounting for both the solid wall and the "ghost" obstacles at $x = 1$, is:

$$P_{t+1}(0) = (q + r)P_t(0) + qP_t(1). \quad (63)$$

Substitute the separation of variables ansatz $P_t(x) = \phi(x)\mu^t$ into this boundary equation:

$$\phi(0)\mu^{t+1} = (q + r)\phi(0)\mu^t + q\phi(1)\mu^t. \quad (64)$$

Divide the entire equation by μ^t to eliminate time:

$$\mu\phi(0) = (q + r)\phi(0) + q\phi(1). \quad (65)$$

7.2 The Cancellation of the Obstacle Density r

In Section 4, we derived the exact temporal eigenvalue including the obstacle "stay" probability r :

$$\mu = 2\sqrt{pq}\cos\theta + r. \quad (66)$$

Let us substitute this full expression for μ into the left side of our boundary equation:

$$(2\sqrt{pq}\cos\theta + r)\phi(0) = (q + r)\phi(0) + q\phi(1). \quad (67)$$

Distribute $\phi(0)$ on the left side:

$$2\sqrt{pq}\cos\theta\phi(0) + r\phi(0) = q\phi(0) + r\phi(0) + q\phi(1). \quad (68)$$

Algebraic Step 18 (The Mathematical Miracle). *Notice that the term $+r\phi(0)$ appears on both sides of the equation. We can subtract $r\phi(0)$ from both sides, causing the obstacle density r to completely vanish from the spatial boundary condition!*

$$2\sqrt{pq}\cos\theta\phi(0) = q\phi(0) + q\phi(1). \quad (69)$$

Intuition 7. *This is a profound mathematical result. It proves rigorously that the environmental obstacles (represented by r) do not alter the spatial shape or the resonant frequencies of the probability wave. The obstacles merely act as a uniform temporal friction (slowing down the decay rate μ by exactly $+r$), but the spatial quantization condition depends only on the spatial hopping bias p and q .*

7.3 Evaluating the Spatial Terms

We now evaluate $\phi(0)$ and $\phi(1)$ using our general spatial form $\phi(x) = R^x \sin((N - x)\theta)$, where $R = \sqrt{p/q}$.

Algebraic Step 19 (Evaluating at $x = 0$).

$$\phi(0) = R^0 \sin((N - 0)\theta) = 1 \cdot \sin(N\theta) = \sin(N\theta). \quad (70)$$

Algebraic Step 20 (Evaluating at $x = 1$).

$$\phi(1) = R^1 \sin((N - 1)\theta) = R \sin((N - 1)\theta). \quad (71)$$

7.4 The Trigonometric Expansion

Substitute $\phi(0)$ and $\phi(1)$ back into the simplified boundary equation $2\sqrt{pq}\cos\theta\phi(0) = q\phi(0) + q\phi(1)$:

$$2\sqrt{pq}\cos\theta\sin(N\theta) = q\sin(N\theta) + q\left[R\sin((N-1)\theta)\right]. \quad (72)$$

Algebraic Step 21 (Isolating the spatial bias R). *Divide the entire equation by q :*

$$2\frac{\sqrt{pq}}{q}\cos\theta\sin(N\theta) = \sin(N\theta) + R\sin((N-1)\theta). \quad (73)$$

Simplify the fraction on the left side: $\frac{\sqrt{pq}}{q} = \sqrt{\frac{p}{q}} = R$.

$$2R\cos\theta\sin(N\theta) = \sin(N\theta) + R\sin((N-1)\theta). \quad (74)$$

Algebraic Step 22 (Applying the Angle Subtraction Identity). *We must expand $\sin((N-1)\theta)$. Using the standard trigonometric identity $\sin(A-B) = \sin A\cos B - \cos A\sin B$, with $A = N\theta$ and $B = \theta$:*

$$\sin((N-1)\theta) = \sin(N\theta)\cos\theta - \cos(N\theta)\sin\theta. \quad (75)$$

Substitute this expansion back into the right side of our equation:

$$2R\cos\theta\sin(N\theta) = \sin(N\theta) + R\left[\sin(N\theta)\cos\theta - \cos(N\theta)\sin\theta\right]. \quad (76)$$

Algebraic Step 23 (Distributing and Consolidating). *Distribute the R on the right side:*

$$2R\cos\theta\sin(N\theta) = \sin(N\theta) + R\sin(N\theta)\cos\theta - R\cos(N\theta)\sin\theta. \quad (77)$$

Subtract $R\sin(N\theta)\cos\theta$ from both sides to consolidate the terms containing $R\cos\theta$:

$$R\cos\theta\sin(N\theta) = \sin(N\theta) - R\cos(N\theta)\sin\theta. \quad (78)$$

Algebraic Step 24 (Grouping the R terms). *Move the remaining R term from the right side to the left side by adding $R\cos(N\theta)\sin\theta$ to both sides:*

$$R\cos\theta\sin(N\theta) + R\sin\theta\cos(N\theta) = \sin(N\theta). \quad (79)$$

Factor out the common R on the left side:

$$R\left[\sin(N\theta)\cos\theta + \cos(N\theta)\sin\theta\right] = \sin(N\theta). \quad (80)$$

Algebraic Step 25 (Applying the Angle Addition Identity). *Recognize the expression inside the brackets as the exact angle addition identity for sine, $\sin(A+B) = \sin A\cos B + \cos A\sin B$, where $A = N\theta$ and $B = \theta$:*

$$\sin(N\theta)\cos\theta + \cos(N\theta)\sin\theta = \sin((N+1)\theta). \quad (81)$$

Substitute this back into the equation:

$$R\sin((N+1)\theta) = \sin(N\theta). \quad (82)$$

7.5 The Transcendental Quantization Condition

Finally, substitute the magnitude $R = \sqrt{p/q}$ back into the equation to obtain the exact quantization condition for the mixed boundary case:

$$\sin(N\theta_n) = \sqrt{\frac{p}{q}} \sin((N+1)\theta_n). \quad (83)$$

This is a **transcendental equation**. Unlike the two-absorbing case, where θ_n was a simple algebraic fraction, the presence of the spatial bias $\sqrt{p/q}$ warps the resonant frequencies. The allowed discrete wavenumbers θ_n (for $n = 1, 2, \dots, N$) are the roots of this equation and must be found numerically.

Intuition 8. *Physically, the reflecting wall does not force the probability wave to zero; it forces the wave to "bounce." Because the random walk is biased ($p \neq q$), the bounce is asymmetric, introducing a complex phase shift. This phase shift prevents a simple algebraic fraction for θ_n . Note: If the walk is perfectly symmetric ($p = q = 1/2$), then $R = 1$, and the equation reduces to $\sin(N\theta) = \sin((N+1)\theta)$, which yields the clean closed-form roots $\theta_n = \frac{\pi(2n-1)}{2N+1}$.*

8 Full Eigenmode Expansion: The Linear Algebra Perspective

8.1 The State Vector and the Transition Matrix

To build deep intuition, let us translate the continuous-index master equation into standard finite-dimensional linear algebra. Let the state of the system at time t be represented by a column vector $\mathbf{P}_t$ of length N , where the x -th entry is the probability $P_t(x)$:

$$\mathbf{P}_t = \begin{bmatrix} P_t(0) \\ P_t(1) \\ \vdots \\ P_t(N-1) \end{bmatrix}. \quad (84)$$

The master equation dictates how this state updates from step t to $t+1$. This update is purely linear and can be written as a matrix multiplication:

$$\mathbf{P}_{t+1} = \mathbf{M}\mathbf{P}_t \implies P_{t+1}(x) = \sum_{y=0}^{N-1} M_{x,y} P_t(y). \quad (85)$$

Here, x is the **row index** (the destination site) and y is the **column index** (the source site). The matrix entry $M_{x,y}$ is the probability of transitioning *from* state y *to* state x .

Because the columns represent the source state y , the sum of each column must account for all probability leaving that state. This makes $\mathbf{M}$ a **column-stochastic matrix**.

For our mixed boundary system (reflecting at 0, absorbing at N , with stay probability r), let us look at a concrete 4×4 example ($N = 4$, valid sites 0, 1, 2, 3, absorber at 4):

$$\mathbf{M} = \begin{bmatrix} q+r & q & 0 & 0 \\ p & r & q & 0 \\ 0 & p & r & q \\ 0 & 0 & p & r \end{bmatrix}. \quad (86)$$

Analyzing the Columns (Source States)

Let us sum the entries in each column to verify probability conservation:

- **Column 0 (Source $y = 0$):** The particle can bounce off the wall/stay ($q + r$) or hop right to site 1 (p). Sum = $(q + r) + p = 1$.
- **Column 1 (Source $y = 1$):** The particle can hop left to 0 (q), stay (r), or hop right to 2 (p). Sum = $q + r + p = 1$.
- **Column 2 (Source $y = 2$):** The particle can hop left to 1 (q), stay (r), or hop right to 3 (p). Sum = $q + r + p = 1$.
- **Column 3 (Source $y = 3$):** The particle can hop left to 2 (q) or stay (r). If it hops right (probability p), it falls into the absorber at $x = 4$ and is lost from the matrix. Sum = $q + r = 1 - p$.

Note 1. Notice that the sum of each column is exactly 1, **except for the last column** (site $N - 1$), which sums to $q + r = 1 - p < 1$. This "missing" probability p is exactly the flux that leaked out into the absorbing boundary at $x = N$.

(Note: If you sum the last **row**, you get $p + r = 1 - q$. However, in a column-stochastic matrix, row sums do not represent probability conservation; only column sums do, because columns track where the probability is coming from).

By applying the matrix recursively, the state at any future time t is simply:

$$\mathbf{P}_t = \mathbf{M}^t \mathbf{P}_0. \quad (87)$$

8.2 Eigendecomposition and Time-Dependent Multipliers

In linear algebra, if we know the eigenvectors $\mathbf{v}_n$ and eigenvalues μ_n of $\mathbf{M}$ (such that $\mathbf{M}\mathbf{v}_n = \mu_n\mathbf{v}_n$), we can decompose any arbitrary initial vector $\mathbf{P}_0$ into a linear combination of these eigenvectors:

$$\mathbf{P}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_N\mathbf{v}_N = \sum_{n=1}^N c_n\mathbf{v}_n. \quad (88)$$

Here, c_n are the *time-independent* scalar coefficients that define the initial "recipe" of eigenvectors.

Now, let us apply the time evolution operator $\mathbf{M}^t$ to this decomposition:

$$\mathbf{P}_t = \mathbf{M}^t \left(\sum_{n=1}^N c_n\mathbf{v}_n \right) = \sum_{n=1}^N c_n \mathbf{M}^t \mathbf{v}_n. \quad (89)$$

Because $\mathbf{M}\mathbf{v}_n = \mu_n\mathbf{v}_n$, it follows that $\mathbf{M}^t\mathbf{v}_n = \mu_n^t\mathbf{v}_n$. Substituting this in:

$$\mathbf{P}_t = \sum_{n=1}^N \underbrace{(c_n \mu_n^t)}_{a_n(t)} \mathbf{v}_n. \quad (90)$$

This reveals the exact meaning of the time-dependent scalar multiplier $a_n(t) = c_n \mu_n^t$. The spatial shape of the mode ($\mathbf{v}_n$, or $\phi_n(x)$ in our continuous notation) never changes. Only its scalar amplitude $a_n(t)$ scales by the eigenvalue μ_n at each time step.

8.3 The Geometry of the Space: Metric Tensors and Generalized Inner Products

To understand how to isolate the coefficients c_n , we must step back and look at the fundamental geometry of vector spaces.

The Standard Euclidean Space

In standard linear algebra, the "dot product" (inner product) of two vectors $\mathbf{u}$ and $\mathbf{v}$ in Euclidean space is defined using the identity matrix $\mathbf{I}$ as the metric tensor:

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{I}} = \mathbf{u}^T \mathbf{I} \mathbf{v} = \sum_i u_i v_i. \quad (91)$$

The metric tensor defines the "units" and geometry of the space. In Euclidean space, the basis vectors are orthogonal and have unit length. A fundamental theorem states that eigenvectors of a **symmetric** matrix corresponding to distinct eigenvalues are strictly orthogonal in this Euclidean space ($\mathbf{u}^T \mathbf{I} \mathbf{v} = 0$).

The Problem with Non-Symmetric Matrices

If our random walk is biased ($p \neq q$), the transition matrix $\mathbf{M}$ is **not symmetric**. Because the matrix is asymmetric, its eigenvectors are *not* orthogonal in standard Euclidean space.

Let us look at a concrete 2D example. Suppose we have two eigenvectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (92)$$

If we take their standard Euclidean dot product:

$$\mathbf{u}^T \mathbf{I} \mathbf{v} = (1)(1) + (0)(1) = 1 \neq 0. \quad (93)$$

They are not perpendicular. If we try to use the standard projection formula to isolate our coefficients, the cross-terms will not vanish, and the algebra will collapse.

Redefining the Space with a Custom Metric Tensor

We do not need to change our vectors; the physical state they represent remains exactly the same. Instead, we change the **geometry of the space** by introducing a custom metric tensor $\mathbf{G}$. The generalized inner product becomes:

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{G}} = \mathbf{u}^T \mathbf{G} \mathbf{v}. \quad (94)$$

We can deliberately choose a metric tensor $\mathbf{G}$ that warps the space such that our specific vectors $\mathbf{u}$ and $\mathbf{v}$ become perfectly orthogonal. For example, let us define:

$$\mathbf{G} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}. \quad (95)$$

Now, let us recalculate the inner product in this new, warped space:

$$\mathbf{u}^T \mathbf{G} \mathbf{v} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1)(1) + (-1)(1) = 0. \quad (96)$$

Intuition 9. *The numbers inside the vectors $\mathbf{u}$ and $\mathbf{v}$ did not change, and the physical probabilities they represent are identical. However, by changing the metric tensor $\mathbf{G}$, we changed the "ruler" we use to measure angles. In this new geometry, the vectors are perfectly perpendicular. This allows us to use the projection trick to isolate the coefficients c_n .*

From Discrete Vectors to Continuous Functions

This geometric concept maps perfectly from finite-dimensional vectors to infinite-dimensional function spaces. For two continuous functions $f(x)$ and $g(x)$ to be orthogonal, their inner product must be zero. In standard Euclidean function space, this is defined via integration (the continuous limit of a discrete sum):

$$\langle f, g \rangle = \int f(x)g(x) dx = 0. \quad (97)$$

Just as we introduced a matrix $\mathbf{G}$ to warp the vector space, we introduce a **weight function** $w(x)$ to warp the function space. The generalized inner product for functions becomes:

$$\langle f, g \rangle_w = \int f(x)w(x)g(x) dx. \quad (98)$$

In our discrete 1D lattice, the "integration" is simply a discrete summation over the sites x . Because the lattice sites are independent basis states, they do not mix with each other in the inner product. Therefore, the metric tensor $\mathbf{G}$ is a **diagonal matrix**, and its diagonal entries G_{xx} are exactly the weight function $w(x)$:

$$\langle \phi_m, \phi_n \rangle_w = \sum_{x=0}^{N-1} w(x)\phi_m(x)\phi_n(x). \quad (99)$$

The Mandatory Metric Tensor for Biased Walks

For our biased random walk, there is one specific metric tensor (weight function) that makes the asymmetric transition operator "self-adjoint" (satisfying the condition of detailed balance). That mandatory metric tensor is:

$$w(x) = \left(\frac{q}{p}\right)^x. \quad (100)$$

When we use this specific weight function to define our inner product, the spatial bias of the eigenfunctions perfectly cancels the weight, reducing the inner product to a pure sum of sines. This guarantees that $\langle \phi_m, \phi_n \rangle_w = 0$ for $m \neq n$.

We have successfully warped the geometry of the probability space so that the eigenmodes of our biased walk are perfectly orthogonal, allowing us to isolate the expansion coefficients c_n via projection.

8.4 Deriving the Weight Function via Detailed Balance

Before applying the weighted inner product, we must answer a fundamental question: *Why is the weight function exactly $w(x) = (q/p)^x$?* Where does this specific mathematical "ruler" come from?

To isolate the coefficients c_n , we need the eigenfunctions to be orthogonal. In linear algebra, eigenvectors are guaranteed to be orthogonal if the matrix operating on them is **symmetric** (or more precisely, self-adjoint with respect to the chosen inner product).

Let $\mathbf{M}$ be our transition matrix, where the entry $M_{x,y}$ represents the probability of transitioning *from* source site y *to* destination site x . For our biased random walk, the matrix is asymmetric because the probability of hopping right (p) differs from hopping left (q):

- Transition from $y = x$ to $x + 1$ (hopping right): $M_{x+1,x} = p$.
- Transition from $y = x + 1$ to x (hopping left): $M_{x,x+1} = q$.

Since $p \neq q$, $M_{x+1,x} \neq M_{x,x+1}$, meaning $\mathbf{M} \neq \mathbf{M}^T$.

The Condition for Self-Adjointness

We want to find a diagonal metric tensor (weight matrix) $\mathbf{W}$ with diagonal entries $w(x)$ such that the operator becomes self-adjoint in this new warped space. The mathematical condition for self-adjointness is that the weighted inner product satisfies $\langle \mathbf{u}, \mathbf{M}\mathbf{v} \rangle_w = \langle \mathbf{M}\mathbf{u}, \mathbf{v} \rangle_w$ for all vectors $\mathbf{u}, \mathbf{v}$.

In terms of matrix elements, this requires the **Detailed Balance** condition to hold for every pair of connected states x and y :

$$w(x)M_{x,y} = w(y)M_{y,x}. \quad (101)$$

Algebraic Step 26 (Applying Detailed Balance to Adjacent Sites). *Let us apply this condition to two adjacent sites, x and $x + 1$. We set $y = x + 1$:*

$$w(x)M_{x,x+1} = w(x+1)M_{x+1,x}. \quad (102)$$

Substitute the physical transition probabilities ($M_{x,x+1} = q$ and $M_{x+1,x} = p$):

$$w(x) \cdot q = w(x+1) \cdot p. \quad (103)$$

Algebraic Step 27 (Solving the Recurrence Relation). *Rearrange the equation to find the ratio of the weights at adjacent sites:*

$$\frac{w(x+1)}{w(x)} = \frac{q}{p}. \quad (104)$$

This is a simple first-order linear recurrence relation (a geometric progression). It states that the weight at any site is exactly the weight of the previous site multiplied by the bias ratio q/p . Solving this recurrence yields:

$$w(x) = w(0) \left(\frac{q}{p} \right)^x. \quad (105)$$

Algebraic Step 28 (Setting the Baseline). *Because the overall scale of the metric tensor does not affect orthogonality (it merely scales the final norm $\mathcal{N}_n$ by a constant factor, which cancels out when solving for c_n), we are free to choose the baseline weight at the origin. We set $w(0) = 1$. This yields the exact, mandatory weight function:*

$$w(x) = \left(\frac{q}{p} \right)^x. \quad (106)$$

Intuition 10. *This derivation reveals that the weight function is not a random mathematical trick; it is the exact mathematical manifestation of **Detailed Balance**. In a system with drift ($p > q$), probability naturally "piles up" on the right side. The weight function $w(x) = (q/p)^x$ exponentially penalizes sites to the right and rewards sites to the left. It perfectly counteracts the physical drift, warping the geometry of the space so that the "flow" of probability appears perfectly symmetric and balanced, thereby restoring orthogonality to the eigenfunctions.*

8.5 Translating Back to Continuous Notation

Returning to our continuous spatial notation, the vector $\mathbf{v}_n$ becomes the eigenfunction $\phi_n(x)$, and the initial state $\mathbf{P}_0$ becomes the Dirac delta $\delta_{x,0}$. The weighted projection formula $c_m = \frac{\langle \mathbf{v}_m, \mathbf{P}_0 \rangle_w}{\langle \mathbf{v}_m, \mathbf{v}_m \rangle_w}$ becomes:

$$c_m = \frac{\sum_{x=0}^{N-1} w(x) \phi_m(x) \delta_{x,0}}{\sum_{x=0}^{N-1} w(x) \phi_m(x)^2} = \frac{\phi_m(0)}{\mathcal{N}_m}, \quad (107)$$

where $\mathcal{N}_m$ is the squared norm of the eigenfunction. This perfectly matches the continuous derivation we performed earlier!

9 The Weighted Inner Product and Orthogonality

9.1 Why the Standard Dot Product Fails

In standard linear algebra, if a matrix is symmetric, its eigenvectors are automatically orthogonal under the standard dot product ($\vec{v}_m \cdot \vec{v}_n = 0$ for $m \neq n$).

However, for a biased random walk ($p \neq q$), the transition matrix is **not symmetric**. The probability of hopping right (p) differs from hopping left (q). Because the matrix is asymmetric, the eigenfunctions $\phi_n(x)$ are *not* orthogonal under the standard unweighted sum:

$$\sum_{x=0}^{N-1} \phi_m(x) \phi_n(x) \neq 0 \quad \text{for } m \neq n. \quad (108)$$

If we tried to use the standard dot product to isolate c_n , the cross-terms would not vanish, and the algebra would collapse.

9.2 The Weight Function as a Metric Tensor

To restore orthogonality, we must warp the geometry of the space by introducing a **weight function**:

$$w(x) = \left(\frac{q}{p}\right)^x. \quad (109)$$

This weight function acts as the diagonal entries of a **metric tensor** G . The weighted inner product of two functions is:

$$\langle \phi_m, \phi_n \rangle_w = \sum_{x=0}^{N-1} w(x) \phi_m(x) \phi_n(x) = \sum_{x=0}^{N-1} \left(\frac{q}{p}\right)^x \phi_m(x) \phi_n(x). \quad (110)$$

Algebraic Step 29 (The Bias Cancellation Miracle). *Substitute the explicit form of the eigenfunctions into the weighted inner product. Recall that $\phi_n(x) = \left(\frac{p}{q}\right)^{x/2} \sin((N-x)\theta_n)$.*

First, multiply the two eigenfunctions together:

$$\phi_m(x)\phi_n(x) = \left(\frac{p}{q}\right)^{x/2} \sin((N-x)\theta_m) \cdot \left(\frac{p}{q}\right)^{x/2} \sin((N-x)\theta_n). \quad (111)$$

Combine the bias factors using the exponent rule $a^{x/2} \cdot a^{x/2} = a^x$:

$$\phi_m(x)\phi_n(x) = \left(\frac{p}{q}\right)^x \sin((N-x)\theta_m) \sin((N-x)\theta_n). \quad (112)$$

Now, multiply by the weight function $w(x) = (q/p)^x$:

$$w(x)\phi_m(x)\phi_n(x) = \underbrace{\left(\frac{q}{p}\right)^x}_{\text{weight}} \cdot \underbrace{\left(\frac{p}{q}\right)^x}_{\text{bias}} \cdot \sin((N-x)\theta_m) \sin((N-x)\theta_n). \quad (113)$$

Since $(q/p)^x \cdot (p/q)^x = 1$ for every site x , the bias completely cancels:

$$w(x)\phi_m(x)\phi_n(x) = \sin((N-x)\theta_m) \sin((N-x)\theta_n). \quad (114)$$

Intuition 11. *The weight function is not arbitrary. It is the exact mathematical object that counteracts the physical drift of the random walk. By warping the geometry of the space with $w(x) = (q/p)^x$, we make the biased transition operator "self-adjoint," guaranteeing that its eigenfunctions are orthogonal.*

9.3 The Orthogonality Relation

With the bias cancelled, the weighted inner product reduces to a pure sum of sines:

$$\langle \phi_m, \phi_n \rangle_w = \sum_{x=0}^{N-1} \sin((N-x)\theta_m) \sin((N-x)\theta_n). \quad (115)$$

Let us change the summation index. Define $j = N - x$. When $x = 0$, $j = N$. When $x = N - 1$, $j = 1$. The sum becomes:

$$\langle \phi_m, \phi_n \rangle_w = \sum_{j=1}^N \sin(j\theta_m) \sin(j\theta_n). \quad (116)$$

For $m \neq n$, this is the standard discrete sine orthogonality relation, which evaluates to zero. For $m = n$, it evaluates to the squared norm $\mathcal{N}_n$. Thus:

$$\langle \phi_m, \phi_n \rangle_w = \mathcal{N}_n \delta_{mn}, \quad (117)$$

where δ_{mn} is the Kronecker delta (1 if $m = n$, 0 if $m \neq n$).

What is Discrete Sine Orthogonality?

In continuous calculus, sine waves of different frequencies are orthogonal over a full period (their integral is zero). In discrete mathematics, we do not have integrals; we have finite sums over a grid of points. **Discrete sine orthogonality** is the property that if you sample sine waves at specific, equally spaced discrete points and multiply them together, their sum is exactly zero, provided the frequencies are distinct and satisfy the boundary conditions. It is the mathematical engine that allows us to separate mixed signals back into their independent components.

To prove this, we use the product-to-sum trigonometric identity $\sin(A)\sin(B) = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$:

$$\langle \phi_m, \phi_n \rangle_w = \frac{1}{2} \sum_{j=1}^N \cos(j(\theta_m - \theta_n)) - \frac{1}{2} \sum_{j=1}^N \cos(j(\theta_m + \theta_n)). \quad (118)$$

The Kronecker Delta ($m \neq n$)

When $m \neq n$, the angles $(\theta_m - \theta_n)$ and $(\theta_m + \theta_n)$ are non-zero multiples of the base frequency. Summing these cosines over the symmetric discrete grid results in perfect destructive interference—the positive and negative areas of the cosine waves exactly cancel each other out, yielding 0. This is represented by the **Kronecker delta** δ_{mn} , which is defined as 0 when $m \neq n$ and 1 when $m = n$. Thus, different eigenmodes are perfectly orthogonal (perpendicular) in our warped space.

The Squared Norm ($m = n$)

When $m = n$, we are taking the inner product of an eigenfunction with itself. In linear algebra, the dot product of a vector with itself is its "length squared" (or squared magnitude). We call this the **Squared Norm**, denoted as $\mathcal{N}_n$. It tells us how "large" or "energetic" that specific eigenmode is in our metric space.

10 Deriving the Expansion Coefficients c_n

10.1 The Projection Trick

To isolate a specific coefficient c_m , we take the weighted inner product of *both sides* of the initial condition equation with $\phi_m(x)$:

$$\sum_{x=0}^{N-1} w(x) \delta_{x,0} \phi_m(x) = \sum_{x=0}^{N-1} w(x) \left[\sum_{n=1}^N c_n \phi_n(x) \right] \phi_m(x). \quad (119)$$

Algebraic Step 30 (Evaluating the Right-Hand Side). *Swap the order of the sums on the right side (finite sums commute):*

$$RHS = \sum_{n=1}^N c_n \underbrace{\left[\sum_{x=0}^{N-1} w(x) \phi_n(x) \phi_m(x) \right]}_{\langle \phi_n, \phi_m \rangle_w}. \quad (120)$$

By orthogonality, the inner sum is $\mathcal{N}_n \delta_{nm}$. Substitute this in:

$$RHS = \sum_{n=1}^N c_n \mathcal{N}_n \delta_{nm}. \quad (121)$$

The Kronecker delta δ_{nm} acts as a filter. It is zero for every term where $n \neq m$. The only term that survives is $n = m$ (where $\delta_{mm} = 1$):

$$RHS = c_m \mathcal{N}_m. \quad (122)$$

Algebraic Step 31 (Evaluating the Left-Hand Side). On the left side, the Dirac delta $\delta_{x,0}$ is a mathematical filter. It equals 1 when $x = 0$ and 0 for all other x . Therefore, the entire sum collapses to just the $x = 0$ term:

$$LHS = w(0) \cdot \phi_m(0) \cdot 1. \quad (123)$$

Evaluate each piece at $x = 0$:

- $w(0) = (q/p)^0 = 1$.
- $\phi_m(0) = \left(\frac{p}{q}\right)^{0/2} \sin((N-0)\theta_m) = 1 \cdot \sin(N\theta_m) = \sin(N\theta_m)$.

Therefore:

$$LHS = \sin(N\theta_m). \quad (124)$$

Algebraic Step 32 (Solving for c_m). Equate the LHS and RHS:

$$\sin(N\theta_m) = c_m \mathcal{N}_m. \quad (125)$$

Solve for c_m :

$$c_m = \frac{\sin(N\theta_m)}{\mathcal{N}_m}. \quad (126)$$

11 Computing the Squared Norm $\mathcal{N}_n$

The squared norm $\mathcal{N}_n$ is the dot product of the eigenfunction with itself. We must evaluate the following sum exactly:

$$\mathcal{N}_n = \sum_{j=1}^N \sin^2(j\theta_n). \quad (127)$$

Algebraic Step 33 (Applying the Half-Angle Identity). Use the trigonometric identity $\sin^2(u) = \frac{1 - \cos(2u)}{2}$:

$$\mathcal{N}_n = \sum_{j=1}^N \frac{1 - \cos(2j\theta_n)}{2} = \frac{1}{2} \sum_{j=1}^N 1 - \frac{1}{2} \sum_{j=1}^N \cos(2j\theta_n). \quad (128)$$

The first sum is simply $\sum_{j=1}^N 1 = N$:

$$\mathcal{N}_n = \frac{N}{2} - \frac{1}{2} \sum_{j=1}^N \cos(2j\theta_n). \quad (129)$$

Algebraic Step 34 (Evaluating the Cosine Sum via Lagrange's Identity). *We must evaluate $C = \sum_{j=1}^N \cos(2j\theta_n)$. We use the standard Lagrange trigonometric identity:*

$$\sum_{j=1}^N \cos(j\alpha) = \frac{\sin(N\alpha/2)}{\sin(\alpha/2)} \cos\left(\frac{(N+1)\alpha}{2}\right). \quad (130)$$

Set $\alpha = 2\theta_n$:

$$C = \sum_{j=1}^N \cos(2j\theta_n) = \frac{\sin(N\theta_n)}{\sin(\theta_n)} \cos((N+1)\theta_n). \quad (131)$$

Algebraic Step 35 (Final Expression for the Norm). *Substitute the cosine sum back into the norm equation:*

$$\mathcal{N}_n = \frac{N}{2} - \frac{1}{2} \left[\frac{\sin(N\theta_n) \cos((N+1)\theta_n)}{\sin(\theta_n)} \right]. \quad (132)$$

$$\boxed{\mathcal{N}_n = \frac{N}{2} - \frac{\sin(N\theta_n) \cos((N+1)\theta_n)}{2 \sin(\theta_n)}} \quad (133)$$

11.1 The Final Expansion Coefficient

Combining our results, the exact expansion coefficient for the n -th mode is:

$$c_n = \frac{\sin(N\theta_n)}{\frac{N}{2} - \frac{\sin(N\theta_n) \cos((N+1)\theta_n)}{2 \sin(\theta_n)}}. \quad (134)$$

11.2 The Surviving Propagator

Because the master equation is linear, any linear combination of valid eigenmodes is also a valid solution. The general form of the surviving propagator is constructed via superposition:

$$P_t(x) = \sum_{n=1}^N c_n \phi_n(x) \mu_n^t. \quad (135)$$

Substituting the explicit expressions we derived for the mixed boundary case with environmental obstacles:

- **Expansion coefficient:** $c_n = \frac{\sin(N\theta_n)}{\mathcal{N}_n}$
- **Spatial eigenfunction:** $\phi_n(x) = \left(\frac{p}{q}\right)^{x/2} \sin((N-x)\theta_n)$
- **Temporal eigenvalue:** $\mu_n = 2\sqrt{pq} \cos \theta_n + r$

into the general superposition yields the exact analytical solution for the surviving probability distribution:

$$P_t(x) = \sum_{n=1}^N \frac{\sin(N\theta_n)}{\mathcal{N}_n} \left(\frac{p}{q}\right)^{x/2} \sin((N-x)\theta_n) (2\sqrt{pq} \cos \theta_n + r)^t. \quad (136)$$

This equation fully describes the probability of finding the surviving particle at site x at time t for a biased random walk with obstacle density $\rho = r$, bounded by a reflecting wall at $x = 0$ and an absorbing cliff at $x = N$.

12 Survival Probability and First Passage Time

12.1 The Survival Probability Recurrence

The survival probability S_t is the total probability that the particle has not yet been absorbed by time t . It is the sum of the probabilities over all valid, non-absorbing sites:

$$S_t = \sum_{x=0}^{N-1} P_t(x). \quad (137)$$

To find the recurrence relation for S_{t+1} , we sum the master equations over all valid sites. We must separate the left boundary ($x = 0$) from the bulk ($1 \leq x \leq N - 1$):

$$S_{t+1} = P_{t+1}(0) + \sum_{x=1}^{N-1} P_{t+1}(x). \quad (138)$$

Substitute the explicit update rules derived in Section 2:

$$S_{t+1} = \left[(q + r)P_t(0) + qP_t(1) \right] + \sum_{x=1}^{N-1} \left[pP_t(x-1) + qP_t(x+1) + rP_t(x) \right]. \quad (139)$$

Algebraic Step 36 (Evaluating the Bulk Sums). *We evaluate the three bulk sums separately by re-indexing. **Right-hop sum:** Let $y = x - 1$. The limits change to $y \in [0, N - 2]$.*

$$p \sum_{y=0}^{N-2} P_t(y) = p \left(S_t - P_t(N-1) \right). \quad (140)$$

Left-hop sum: Let $y = x + 1$. The limits change to $y \in [2, N]$. Since $P_t(N) = 0$:

$$q \sum_{y=2}^N P_t(y) = q \left(S_t - P_t(0) - P_t(1) \right). \quad (141)$$

Stay sum: The sum over the bulk sites is simply the total sum minus the $x = 0$ site:

$$r \sum_{x=1}^{N-1} P_t(x) = r \left(S_t - P_t(0) \right). \quad (142)$$

Algebraic Step 37 (Assembling and Cancelling Terms). *Substitute these evaluated sums back into the equation for S_{t+1} :*

$$S_{t+1} = \underbrace{qP_t(0) + rP_t(0) + qP_t(1)}_{\text{from } x=0} \quad (143)$$

$$+ \underbrace{pS_t - pP_t(N-1)}_{\text{right-hops}} \quad (144)$$

$$+ \underbrace{qS_t - qP_t(0) - qP_t(1)}_{\text{left-hops}} \quad (145)$$

$$+ \underbrace{rS_t - rP_t(0)}_{\text{stays}}. \quad (146)$$

Group the S_t terms: $(p + q + r)S_t = (1)S_t = S_t$. Group the $P_t(0)$ terms: $qP_t(0) + rP_t(0) - qP_t(0) - rP_t(0) = 0$. Group the $P_t(1)$ terms: $qP_t(1) - qP_t(1) = 0$.

Algebraic Step 38 (The Mathematical Miracle). *All boundary terms on the left side perfectly cancel out! We are left with:*

$$S_{t+1} = S_t - pP_t(N-1). \quad (147)$$

Intuition 12. *This algebraic cancellation is the mathematical signature of a true reflecting wall. Even with the addition of the obstacle "stay" probability r , the probability that attempts to leave the left boundary (either by hitting the solid wall or hitting an obstacle at $x = 1$) is instantly returned to the system. No probability leaks from the left; the only net loss is the flux hopping from $N-1$ to N .*

12.2 First Passage Time (FPT) Distribution

The First Passage Time F_t is the probability that the particle is absorbed at exactly step t . By definition, this is the amount of survival probability lost between step $t-1$ and step t :

$$F_t = S_{t-1} - S_t. \quad (148)$$

Using our survival recurrence (shifted to $t-1$), we immediately obtain:

$$F_t = pP_{t-1}(N-1). \quad (149)$$

To find the explicit closed-form solution, we substitute the eigenmode expansion $P_{t-1}(N-1) = \sum_{n=1}^N c_n \phi_n(N-1) \mu_n^{t-1}$. Evaluate the eigenfunction at the boundary $x = N-1$:

$$\phi_n(N-1) = \left(\frac{p}{q}\right)^{\frac{N-1}{2}} \sin((N - (N-1))\theta_n) = \left(\frac{p}{q}\right)^{\frac{N-1}{2}} \sin(\theta_n). \quad (150)$$

Substituting this and the coefficient $c_n = \frac{\sin(N\theta_n)}{\mathcal{N}_n}$ into the FPT equation yields the exact, closed-form FPT distribution:

$$F_t = p \left(\frac{p}{q}\right)^{\frac{N-1}{2}} \sum_{n=1}^N \frac{\sin(N\theta_n) \sin(\theta_n)}{\mathcal{N}_n} \mu_n^{t-1}, \quad t \geq 1. \quad (151)$$

13 Mean First Passage Time (MFPT)

While the MFPT $\langle T \rangle = \sum_{t=1}^{\infty} t F_t$ can be found by summing the infinite series, it is vastly more efficient to solve the backward Kolmogorov difference equation. Let T_x be the expected number of time steps to reach the absorbing boundary starting from site x .

13.1 The Backward Equation with Obstacles

At each time step, 1 unit of time passes. The particle then hops right with probability p , left with probability q , or stays in place with probability r . The recurrence for the bulk is:

$$T_x = 1 + pT_{x+1} + qT_{x-1} + rT_x. \quad (152)$$

Move rT_x to the left side:

$$(1-r)T_x = 1 + pT_{x+1} + qT_{x-1}. \quad (153)$$

Since $p + q + r = 1$, we know that $1-r = p+q$. Substitute this in:

$$(p+q)T_x - pT_{x+1} - qT_{x-1} = 1. \quad (154)$$

Intuition 13. Notice that the obstacle probability r has completely vanished from the spatial recurrence! The equation is identical to a random walk without obstacles, but where the spatial hop probabilities p and q do not sum to 1. The term $\frac{1}{p+q} = \frac{1}{1-r}$ represents the expected number of time steps the particle must wait at site x before successfully making a spatial hop.

13.2 Boundary Conditions

Absorbing Boundary ($x = N$): The walk ends immediately, so $T_N = 0$. **Reflecting Boundary** ($x = 0$): The update rule is $T_0 = 1 + pT_1 + qT_0 + rT_0$. Move qT_0 and rT_0 to the left side:

$$(1 - q - r)T_0 = 1 + pT_1. \quad (155)$$

Since $1 - q - r = p$, this simplifies beautifully to:

$$pT_0 = 1 + pT_1 \implies T_0 - T_1 = \frac{1}{p}. \quad (156)$$

This is exactly the same boundary condition as the $r = 0$ case!

13.3 Solving the Difference Equation

The general solution to the non-homogeneous difference equation $(p + q)T_x - pT_{x+1} - qT_{x-1} = 1$ is the sum of the homogeneous solution and a particular linear solution $T_x = Cx$:

$$T_x = A + B \left(\frac{q}{p} \right)^x + \frac{x}{q - p}. \quad (157)$$

Algebraic Step 39 (Applying the Boundary Conditions). Apply $T_N = 0$:

$$A + B \left(\frac{q}{p} \right)^N + \frac{N}{q - p} = 0 \implies A = -B \left(\frac{q}{p} \right)^N - \frac{N}{q - p}. \quad (158)$$

Apply $T_0 - T_1 = 1/p$:

$$T_0 = A + B, \quad (159)$$

$$T_1 = A + B \left(\frac{q}{p} \right) + \frac{1}{q - p}. \quad (160)$$

Subtracting T_1 from T_0 :

$$T_0 - T_1 = B \left(1 - \frac{q}{p} \right) - \frac{1}{q - p} = B \left(\frac{p - q}{p} \right) + \frac{1}{p - q}. \quad (161)$$

Set this equal to $1/p$:

$$B \left(\frac{p - q}{p} \right) + \frac{1}{p - q} = \frac{1}{p} \implies B \left(\frac{p - q}{p} \right) = \frac{1}{p} - \frac{1}{p - q} = \frac{-q}{p(p - q)}. \quad (162)$$

Solving for B :

$$B = \frac{-q}{(p - q)^2}. \quad (163)$$

Algebraic Step 40 (Evaluating the MFPT at $x = 0$). We seek $T_0 = A + B$. Substitute A :

$$T_0 = -B \left(\frac{q}{p} \right)^N - \frac{N}{q-p} + B = B \left[1 - \left(\frac{q}{p} \right)^N \right] + \frac{N}{p-q}. \quad (164)$$

Substitute $B = \frac{-q}{(p-q)^2}$:

$$\langle T \rangle = T_0 = \frac{N}{p-q} - \frac{q}{(p-q)^2} \left[1 - \left(\frac{q}{p} \right)^N \right]. \quad (165)$$

13.4 The Physical Meaning of Obstacles on MFPT

The formula above is expressed in terms of the *effective* probabilities p and q . Let us rewrite it in terms of the *intrinsic* probabilities p_0, q_0 and the obstacle density ρ . Recall that $p = p_0(1 - \rho)$, $q = q_0(1 - \rho)$, and $r = \rho$. Therefore, $p - q = (p_0 - q_0)(1 - \rho)$, and the ratio $q/p = q_0/p_0$. Substitute these into the MFPT formula:

$$\langle T \rangle = \frac{N}{(p_0 - q_0)(1 - \rho)} - \frac{q_0(1 - \rho)}{(p_0 - q_0)^2(1 - \rho)^2} \left[1 - \left(\frac{q_0}{p_0} \right)^N \right]. \quad (166)$$

Factor out $\frac{1}{1-\rho}$ from both terms:

$$\langle T \rangle = \frac{1}{1 - \rho} \left(\frac{N}{p_0 - q_0} - \frac{q_0}{(p_0 - q_0)^2} \left[1 - \left(\frac{q_0}{p_0} \right)^N \right] \right). \quad (167)$$

$$\boxed{\langle T \rangle = \frac{1}{1 - \rho} \langle T \rangle_{\text{free}}} \quad (168)$$

where $\langle T \rangle_{\text{free}}$ is the MFPT of a random walk in a completely free environment ($\rho = 0$).

Intuition 14. *This is a spectacular result. The expected time to reach the target in a disordered environment is exactly the expected time in a free environment, multiplied by $\frac{1}{1-\rho}$. Because $1 - \rho$ is the probability of making a successful spatial hop, $\frac{1}{1-\rho}$ is the expected number of time steps wasted "bouncing" off obstacles per successful spatial hop. The obstacles do not change the spatial path the particle takes; they merely insert a geometrically distributed temporal delay at every single site.*

14 Final Conclusion

We have exhaustively derived the exact First Passage Time distribution and Mean First Passage Time for a 1D biased random walk with environmental disorder, bounded by a reflecting wall and an absorbing cliff.

By explicitly marginalizing the environment (the Annealed Approximation), we proved that the obstacle density ρ acts purely as a temporal friction $r = \rho$. This friction shifts the temporal eigenvalues $\mu_n \rightarrow \mu_n + \rho$ and scales the MFPT by $\frac{1}{1-\rho}$, but it leaves the spatial eigenfunctions and the transcendental quantization condition entirely unchanged. This deep mathematical decoupling allows for highly efficient computational validation of autonomous systems navigating stochastic, obstacle-dense environments.